Iran's Nuclear Threshold

JCPOA Collapse, 60% Enrichment, and the One-Week Breakout Estimate

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Executive Summary

Iran has reached a nuclear threshold position through a decades-long enrichment programme that Western states have long accused of seeking weapons capability. By September 2025, on the day Israel launched its strikes, Iran held 440.9 kg of uranium enriched to 60% purity — slightly more than previously reported — and the IAEA noted that amount was "enough in principle" for multiple weapons if further enriched to weapons-grade CNA[2025-09-04]. The 60% level is a short step from the roughly 90% constituting weapons-grade, and far exceeds any civilian nuclear justification CNA[2025-09-04]. The U.S. government estimated in March 2022 that Iran would have needed as little as one week to produce enough weapons-grade HEU for one nuclear weapon; the fissile material stockpile would, if further enriched, have been sufficient for "more than a dozen nuclear weapons" [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)]. The IAEA has separately reported that Iran does not yet have a viable nuclear weapon design or a suitable explosive detonation system, and Tehran may also lack sufficient experience in producing uranium metal in weapons-usable form [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)]. The Joint Comprehensive Plan of Action, signed in 2015 and abandoned by the United States in 2018, has not been replaced by any constraining framework. Iran stands as a de facto nuclear threshold state.

Part I — JCPOA Architecture and Collapse

The Joint Comprehensive Plan of Action was the product of multilateral negotiations between Iran and the P5+1 — the five permanent UN Security Council members plus Germany — with the European Union as coordinator CNA[2025-08-13]. The agreement offered Iran relief from international sanctions in exchange for limits on uranium enrichment and acceptance of international monitoring; all three signatories Britain, France, and Germany were parties alongside the United States, China, and Russia CNA[2025-08-13].

President Trump withdrew the United States from the JCPOA in 2018 and ordered new sanctions CNA[2025-08-13]. Iran's Foreign Minister Abbas Araghchi later argued that the European countries did not have the legal right to restore sanctions under the agreement; the E3 called this allegation "unfounded," asserting they were "clearly and unambiguously legally justified" as JCPOA signatories in triggering UN snapback to reinstate Security Council resolutions that would prohibit enrichment CNA[2025-08-13]. Following the US withdrawal, critics of the deal — including Trump and his administration — described it as "one-sided" and insufficient; Trump transitioned to a "maximum pressure" campaign pairing unilateral sanctions with renewed threats of military force [RAG: MILITARY_RAG — Parameters: INSTRUMENTS OF NATIONAL POWER (2024-01-01)].

Iran began resuming activities exceeding JCPOA limits following the US withdrawal in July 2019 [RAG: MILITARY_RAG — ODNI Annual Threat Assessment 2022 (2022-02-07)]. The ODNI assessed in 2022 that Iran was not at that time undertaking key nuclear weapons-development activities judged necessary to produce a nuclear device, but that Iranian officials would probably consider further enriching uranium up to 90% if they did not receive sanctions relief [RAG: MILITARY_RAG — ODNI Annual Threat Assessment 2022 (2022-02-07)]. The same assessment had been made in 2021: following the 2018 withdrawal, Iranian officials abandoned some JCPOA commitments and resumed nuclear activities beyond JCPOA limits, with Iran again flagging that without sanctions relief it would consider options including further enrichment [RAG: MILITARY_RAG — ODNI Annual Threat Assessment 2021 (2021-04-09)].

In January 2021, Iran began enriching uranium to 20% at the buried Fordow facility and started research and development with the stated intent to produce uranium metal for research reactor fuel; it produced gram quantities of natural uranium metal in a laboratory experiment that same month [RAG: MILITARY_RAG — ODNI Annual Threat Assessment 2021 (2021-04-09)]. The IAEA verified in 2021 that Iran conducted research on uranium metal production and produced small quantities of uranium metal enriched to 20%; the production of uranium metal was prohibited under the JCPOA as a key capability needed to produce nuclear weapons [RAG: MILITARY_RAG — ODNI Annual Threat Assessment 2023 (2023-02-06)].

Iran officially started restricting international inspections of its nuclear facilities on February 23, 2021, in a bid to pressure European countries and the Biden administration to lift sanctions CNA[2021-02-24].

Part II — Current Nuclear Status (2025–2026)

On the day Israel launched its attacks in 2025, Iran held 440.9 kg of uranium enriched to 60% purity — slightly more than previously reported in a confidential IAEA report sent to member states and seen by Reuters CNA[2025-09-04]. The IAEA chief told Reuters on September 3, 2025 that the Agency had had no information from Iran on the status or whereabouts of its nuclear material following the strikes, and that talks on resuming inspections at bombed sites — including Natanz and Fordow — could not continue for months on end CNA[2025-09-04].

Following the strikes, the IAEA demanded Iran open up the bombed nuclear sites, including the Natanz uranium enrichment plant and Fordow underground enrichment complex CNA[2025-11-20]. Iran's response was to cooperate only regarding nuclear facilities that had not been affected CNA[2025-11-20]. The IAEA has not been able to inspect the attacked Iranian nuclear facilities; U.S. government reports provided differing assessments of the June strikes' impacts — the 2025 U.S. National Security Strategy stated the strikes "significantly degraded" Iran's nuclear programme, while the 2026 National Defense Strategy characterised the impact differently [RAG: MILITARY_RAG — CRS: U.S. Military Operations Against Iran's Missile and Nuclear Programs (2026-03-06)].

The 2018 U.S. Nuclear Posture Review noted Iran's continuation of the largest missile programme in the Middle East and assessed that Iran could in the future threaten or deliver nuclear weapons were it to acquire them following JCPOA expiration [RAG: MILITARY_RAG — 2018 Nuclear Posture Review (2018-02-02)]. Subsequent Joint Chiefs Chairman testimony in March 2023 assessed Iran would need "several months to produce an actual nuclear weapon," though this estimate was not fully explained in terms of its underlying assumptions [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)]. The Defense Intelligence Agency assessed in May 2025 that Iran would have needed "prob[ably...]" — the chunk is truncated — to reach weapons capability [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)].

The breakout concept was described by a State Department official in September 2021 as "a useful metric to help quantify" U.S. negotiating goals and "a useful analytic framework," and was used during JCPOA negotiations as an unclassified proxy measure of Iranian nuclear weapons capabilities [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)].

The IAEA had previously noted Natanz as the central facility for advanced uranium enrichment equipment. As of June 2013, the IAEA's quarterly report found that Tehran had accelerated installation of advanced enrichment equipment at its central Natanz plant — a development Iran's envoy described as validation of peaceful activity, while the IAEA report itself noted it opened a potential second route to developing a bomb CNA[2013-06-10]. An earlier IAEA report in November 2011 found intelligence that Iran had conducted research into developing nuclear weapons until 2003 and possibly since then; Iran insisted its programme was civilian CNA[2014-05-24].

Part III — Military Operations Against Iran

U.S. forces struck Iranian nuclear sites in June 2025; this constituted the fifth U.S. military engagement with Iran since the Hamas attack on Israel in October 2023 — the prior four being U.S. defence of Israel during Iran's conflicts with Israel in April 2024, November 2024, and June 2025 [RAG: MILITARY_RAG — CRS: U.S. and Israeli Military Operations Against Iran: Issues for Congress (2026-03-01)]. U.S. and Israeli military operations against Iran marked a fifth engagement with Iran in that period [RAG: MILITARY_RAG — CRS: U.S. and Israeli Military Operations Against Iran: Issues for Congress (2026-03-01)].

Following Iran's unprecedented attacks against Israel on April 13 and October 1, 2024, the Department of Defense deployed a Terminal High-Altitude Area Defense (THAAD) battery and associated U.S. military personnel to Israel to bolster Israeli air defences [RAG: MILITARY_RAG — CRS: The Terminal High Altitude Area Defense (THAAD) System (2026-06-23)]. The United States employed ballistic missile defence systems including PATRIOT, THAAD, and BMD-capable destroyers to counter Iranian ballistic missile threats; the DOD stated its stockpile of PATRIOT interceptors "remains extremely strong" [RAG: MILITARY_RAG — CRS: U.S. Military Operations Against Iran: Munitions and Missile Defense (2026-03-12)].

Subsequent Israeli strikes on Iran after the April and October 2024 ballistic missile attacks "destroyed Iran's ability to produce ballistic missiles for a year," according to U.S. assessment [RAG: MILITARY_RAG — CRS: Iran's Ballistic Missile Programs: Background and Context (2025-06-17)].

President Trump announced on February 28, 2026 that the U.S. military had struck alongside Israel [RAG: MILITARY_RAG — Lowy: Australians register historic levels of distrust and unease in 2026 (2022-01-01)].

Part IV — Snapback Sanctions and the Diplomatic Framework

Britain, France, and Germany launched a 30-day process on August 28, 2025 to reimpose UN sanctions — the snapback mechanism — accusing Tehran of failing to abide by the 2015 deal CNA[2025-09-25]. Any of the current JCPOA members — France, the United Kingdom, Germany, China, the European Union, and Russia — retained the right to trigger snapback to reimpose United Nations sanctions even though the U.S. had withdrawn from the agreement in 2018 CNA[2025-04-09]. Iran once again faced an arms embargo and a ban on uranium enrichment and reprocessing activities after attempts to delay the return of sanctions at the UN failed CNA[2025-09-28]. Iran's envoy to the IAEA, Reza Najafi, said the IAEA resolution demanding access to bombed sites would have a "negative impact" on relations with the UN agency CNA[2025-11-20].

Iran's Supreme Leader stated that enriched uranium must stay in Iran CNA[2026-05-21]. Western states have long accused Iran of seeking nuclear weapons, including pointing to its move to enrich uranium far higher than needed for civilian uses CNA[2026-05-21].

In 2017, the United States, Britain, France, and Germany warned the United Nations that Iran had taken "a threatening and provocative step" by testing a rocket capable of delivering satellites into orbit CNA[2017-08-02].

Part V — Regional and Strategic Assessment

The Atlantic Council's Global Foresight 2025 survey found that 88% of respondents expected at least one new country to acquire nuclear weapons in the coming decade, with Iran identified as the most likely — but not the only — potential new nuclear-weapons power on the horizon [RAG: THINK_TANK_RAG — Atlantic Council: Global Foresight 2025 (2025-01-01)]. The percentage of respondents anticipating a nuclear Iran in ten years remained steady year over year, as did approximately 40% of respondents expecting Saudi Arabia to acquire nuclear weapons as well [RAG: THINK_TANK_RAG — Atlantic Council: Global Foresight 2025 (2025-01-01)].

The Atlantic Council assessed that with Iran's axis of resistance shredded and Iran weakened militarily and economically, the United States had an extraordinary opportunity — working with Israel, Arab allies, and European countries — to use economic and diplomatic pressure backed by the threat of military force to secure an agreement walking Iran back from the nuclear brink and curbing its destabilising regional policies [RAG: THINK_TANK_RAG — Atlantic Council: Global Foresight 2025 (2025-01-01)].

A U.S. National Intelligence Estimate concluded that while Iran likely halted its weapons programme in fall 2003, "Tehran at a minimum is keeping open the option to develop nuclear weapons" in the future; IAEA Director General Mohamed ElBaradei stated in September 2009 that Iran continued to fail to cooperate with the Agency, making it impossible to determine Tehran's intent [RAG: THINK_TANK_RAG — CFR Backgrounder: Iran's Nuclear Program (2024-01-01)]. The IAEA had presented Iran in February 2008 with intelligence collected by the United States concerning possible military dimensions [RAG: THINK_TANK_RAG — CFR Backgrounder: Iran's Nuclear Program (2024-01-01)].

Arms transfers to the Middle East fell 13% between 2016–20 and 2021–25; three of the world's top ten arms recipients in 2021–25 were in the region — Saudi Arabia (rank 3), Qatar (rank 4), and Kuwait (rank 9) — with more than half of Middle East arms imports coming from the United States (54%) [RAG: MILITARY_RAG — SIPRI: Trends in International Arms Transfers, 2025 (2026-01-01)].

Conclusion

Iran's enrichment programme has survived military strikes, sanctions, and diplomatic isolation to produce a 60%-enriched stockpile that the IAEA characterised as sufficient in principle for multiple nuclear weapons if further enriched CNA[2025-09-04]. The IAEA cannot inspect the bombed facilities and has no confirmed information on the current disposition of Iran's fissile material CNA[2025-11-20]. The snapback sanctions mechanism has been triggered and UN restrictions reimposed CNA[2025-09-28], but no diplomatic framework currently constrains enrichment. Iran has no verified viable weapon design or detonation system as of available reporting [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)], yet the fissile material stockpile was assessed as sufficient for more than a dozen weapons if further enriched, with a one-week breakout timeline for a single weapon's worth of HEU [RAG: MILITARY_RAG — CRS: Iran and Nuclear Weapons Production (2026-04-09)]. The structural gap between what Iran requires and what the international community has been willing to offer remains unbridged.

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